

(1.1) Ako skáľaron čísl máme:

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle$$

$$\text{tak } \|u+v\|^2 - \|u-v\|^2 = 2\langle u, v \rangle + 2\langle v, u \rangle = 4(\langle u, v \rangle + \overline{\langle u, v \rangle}) = 4\operatorname{Re}(\langle u, v \rangle).$$

$$\text{tedy } \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2).$$

Pro imaginárni časť máme:

$$\|u-iv\|^2 = \langle u-iv, u-iv \rangle = \langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \langle v, v \rangle$$

$$\|u+iv\|^2 = \langle u+iv, u+iv \rangle = \langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \langle v, v \rangle$$

$$\text{tak } \|u-iv\|^2 - \|u+iv\|^2 = -2i\langle u, v \rangle + 2i\langle v, u \rangle = 2i(\overline{\langle u, v \rangle} - \langle u, v \rangle)$$

$$\text{a keďže je } \bar{z} - z = -2i\operatorname{Im}(z), \text{ tak máme } \|u-iv\|^2 - \|u+iv\|^2 = 4\operatorname{Im}(\langle u, v \rangle).$$

$$\text{tedy } \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2)$$